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A NEW PALAEOMAGNETIC RESULT FROM EAST AFRICA AND ESTIMATES OF THE MESOZOIC PALAEORADIUS

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Of 16 sites collected in the Taru grits (Permian) and Maji ya Chumvi beds (Permo-Triassic) of East Africa only 6 sites from the Maji ya Chumvi sediments gave meaningful palaeomagnetic results. After thermal cleaning the 6 sites (32 samples) give an Early Triassic pole at 67°N, 269°E with $A_{95} = 17°$ in excellent agreement with other African Mesozoic poles. There are now 26 Mesozoic palaeomagnetic poles for Africa from widely diverse localities ranging in present latitude from 35°N to 30°S. The poles subdivide into Triassic (17 poles) and Cretaceous (9 poles) groups whose means are not significantly different. The palaeomagnetic pole for Africa thus remained in much the same position for 170 m.y. from Early Triassic to Late Cretaceous. The data form an especially good set for estimating the palaeoradius using Ward's method. Values of 1.08 ± 0.15 and 1.03 ± 0.19 times the present radius are deduced for the Triassic and Cretaceous respectively with a mean value of 1.08 ± 0.13 for all the Mesozoic data combined. The analysis demonstrates that hypotheses of earth expansion are very unattractive.

1. Introduction

Early work on the palaeomagnetism of Mesozoic rocks in Africa was restricted to detailed observations of rocks of the Karroo Supergroup in southern Africa [1,2]. The data showed that during most of the Mesozoic at least up to mid-Cretaceous times the palaeomagnetic pole remained stationary with respect to Africa. These measurements have since been supplemented by extensive work in northwest Africa [3–7] which both confirm and strengthen the work in southern Africa. More recently work in Egypt [8] has shown that the constant Mesozoic pole position can be extended in time to include also the Late Cretaceous. Agreement of these latter poles with data from Sicily of the same age has shown that the island formed part of the African plate at least as far back as the Cretaceous [9]. In this paper we present the first results from the Karroo Supergroup in Kenya involving partial demagnetization cleaning procedures. They show that the constancy of the Mesozoic pole position with respect to Africa extended back to the Palaeozoic/Mesozoic boundary.

2. Geology and sampling

A strip about 60 km in width parallel with and near to the coast in Kenya is occupied by sediments of the Karroo Supergroup. The sediments are known as the Duruma Sandstones and are subdivided into the Taru grits (Permian) at the base overlain successively by the Maji ya Chumvi beds, the Mariakani Sandstone and the Mazeras Sandstone which together range in age from near the Permo-Triassic boundary through the Triassic and into the Early Jurassic. Eighty four cores were col-

lected at sixteen sites in the Taru grits and Maji ya Chumvi formation along the main road between Nairobi and Mombasa in the Mackinnon Road–Mariakani area (3.8°S, 39.4°E). Cores 22 mm in diameter were collected using a portable sampling drill and were oriented using a prismatic compass. A reconnaissance survey of these rocks without the use of cleaning techniques has previously been made by Nairn [10].

The geology of the area has been described by Miller [11]. The Taru grits are a fluviatile formation derived directly from the basement rocks further west. It is divided into a lower division of sandy shales, arkoses and conglomerates and an upper division of well bedded yellow sandstones and arkoses with interbedded conglomerate lenticles. Ten sites were collected (52 cores) in the Taru grits, six from the lower division and four from the upper division. The Maji ya Chumvi beds are probably slightly unconformable to the Taru grits and mark a change to quiet lagoonal sedimentation. Yellow rippled sandstones overlie the grits, passing by alternation into thin-bedded carbonaceous muddy sandstones and gritty shales in which fossil plants of Late Permian to Early Triassic age have been found. These are abruptly succeeded by soft blue shales containing Early Triassic marine fish remains. This marine horizon within the Maji ya Chumvi formation and which separates the lower beds from the upper beds is particularly relevant in correlation with Madagascar where a similar horizon occurs within the Sakamena beds [12]. One site was located in the lower Maji ya Chumvi beds and five sites in the upper beds (total 32 cores). These sites are almost certainly confined within the Early Triassic although the one site in the lower beds could just lie at the top of the Permian. The beds have a general dip of 5° ESE.

3. Palaeomagnetic results

Measurements of the natural remanent magnetization (NRM) were initially made during 1971 using a PAR-SM1 spinner magnetometer. However, most intensities were less than 1.0 mA m^{-1} (10^{-6} emu cm^{-3}) and generally in the range 0.1–0.5 mA m^{-1}. This is at the lower limit of measurement on the magnetometer so that no demagnetization tests could be carried out. Further work was delayed for three years until a Digico Complete Results magnetometer became available. No difficulties were encountered with this mag-

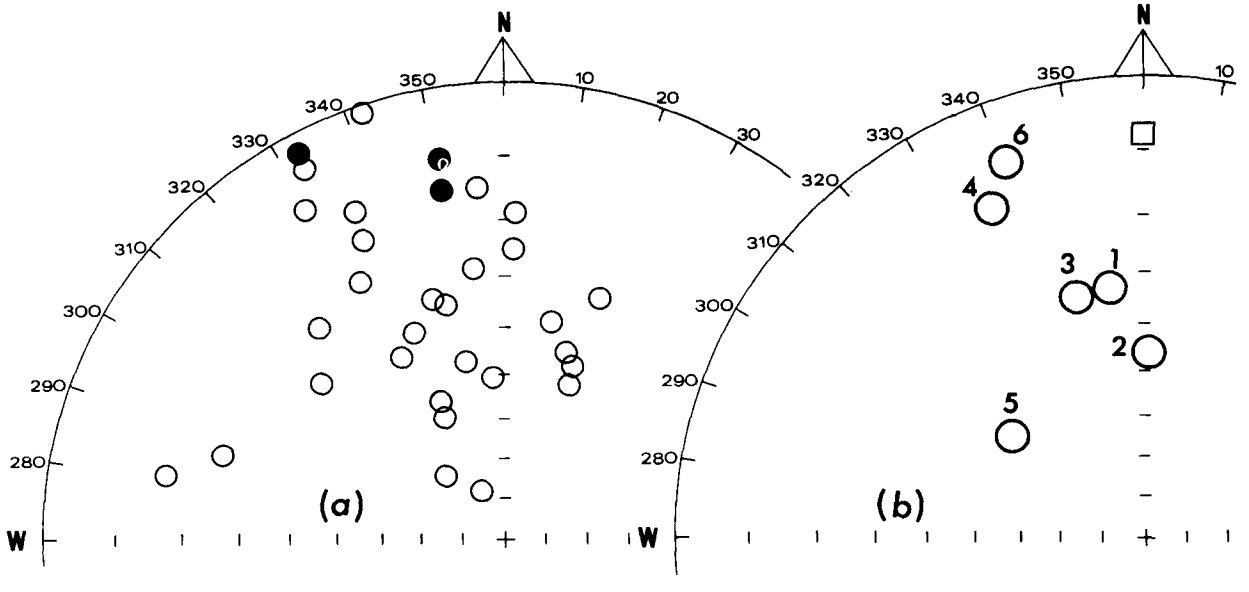


Fig. 1. Stereographic projection of directions of magnetization observed in the Maji ya Chumvi beds after thermal cleaning at 300°C. Solid symbols, lower hemisphere; open symbols, upper hemisphere. The square is the axial dipole field direction. (a) Sample directions. (b) Site mean directions.

TABLE 1

Palaeomagnetic results from the Maji ya Chumvi formation (3.8°S, 39.4°E)*

Site	N	NRM				Cleaned at 300°C				VGP	
		D	I	R	k	D	I	R	k	lat. °N.	long. °E
1	5	336	−11	4.665	11.9	354	−32	4.700	13.3	75	244
2	5	331	−37	4.524	8.4	2	−46	4.792	19.1	66	214
3	6	345	−43	5.089	5.4	345	−33	5.455	9.1	69	264
4	6	353	−10	5.317	7.3	336	−14	5.530	10.6	66	300
5	6	333	−60	5.077	5.4	309	−51	5.331	7.4	35	273
6	4	339	−5	3.713	10.4	340	−8	3.857	21.0	70	307
Mean	6	336	−27	5.536	10.7	342	−32	5.643	13.9	67	269 (A_{95} = 17°)

* D, I are the declination (clockwise from true north) and inclination (negative upwards), R the length of the resultant of N unit vectors and k the Fisher [27] precision parameter. A_{95} is the radius of the circle of 95% confidence about the mean pole position.

netometer which measured intensities down to 0.05 mA m^{-1} or better.

Pilot samples from each of the sixteen sites were thermally demagnetized at increasing temperatures using standard laboratory equipment [13]. Samples from the Taru grits were initially widely scattered and thermal treatment was unable to produce any meaningful results. However, samples from the Maji ya Chumvi beds responded to thermal cleaning, the best grouping of directions being observed at 300°C, when intensities had dropped to 0.1 mA m^{-1} or less. At higher temperatures some samples became too weakly magnetized for meaningful measurement, but the remainder showed stability up to 500 or 600°C above which temperature none of the samples was measureable.

Fig. 1 shows sample and site mean directions after thermal cleaning. All samples were normally magnetized. The various statistical parameters and pole position derived from the study are given in Table 1. The palaeomagnetic field during the Early Triassic is known to contain many polarity changes [21,22] so it is surprising that only normal polarity has been observed in the Maji ya Chumvi beds. This may indicate that the horizons sampled cover only a narrow time range, a possibility that remains unresolved because of the poor exposure of these rocks. However, the pole position is in excellent agreement with other Mesozoic poles for Africa (Fig. 2, Table 2). Since these rocks have an

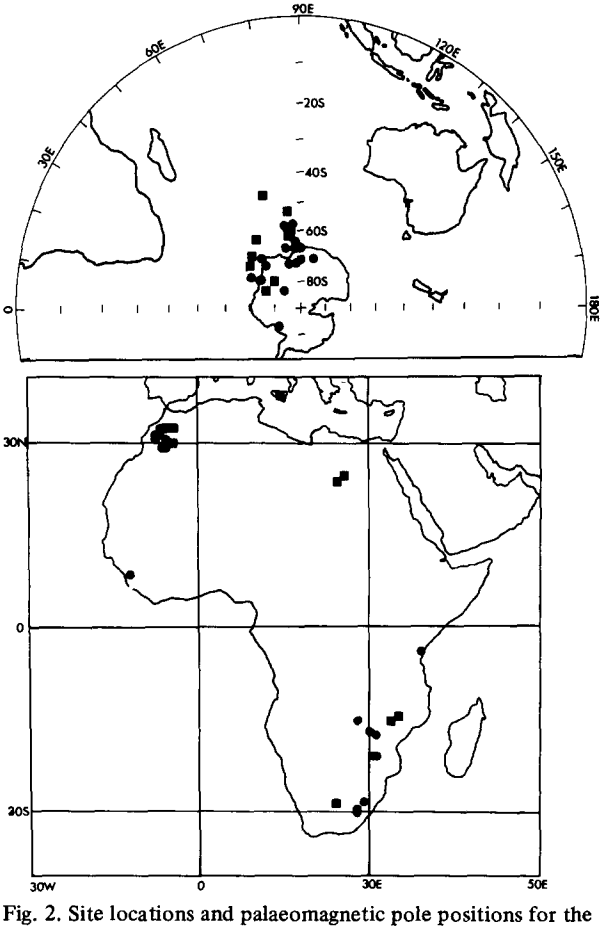


Fig. 2. Site locations and palaeomagnetic pole positions for the Mesozoic of Africa. Circles, Triassic; squares, Cretaceous.

TABLE 2
Summary of Mesozoic palaeomagnetic poles for Africa*

Rock unit	Age	Mean location	Pole position	Reference
Maji ya Chumvi beds	Trl	39.5°E, 4°S	269°E, 67°N	this paper
Beaufort sediments	Trl	28°E, 30°S	267°E, 67°N	[14]
Issaldain volcanics	Trm	352.5°E, 31°N	140.5°E, 77.5°N	[7]
Ait-Aadel volcanics	Trm	352.5°E, 31.5°N	254.5°E, 72°N	[7]
Titchka sediments	Trm	352.5°E, 31.5°N	267°E, 68.5°N	[7]
Shawa ijolite	Tru	31.5°E, 19°S	265.5°E, 64°N	[15]
Red sandstone formation	Tru	28.5°E, 16°S	229.5°E, 68°N	[16]
Marangudzi Complex	Tru	30.5°E, 22°S	285°E, 70°N	[18]
Stormberg lavas	Tru-Jl	28.5°E, 29.5°S	269°E, 71°N	[2]
Karroo lavas, Central Africa	Tru-Jl	30°E, 18°S	264°E, 57°N	[2]
Karroo dolerites	Tru-Jl	28°E, 30°S	255°E, 65.5°N	[2]
Moroccan intrusives	Tru-Jl	352.5°E, 32°N	216°E, 71°N	[5]
Draa Valley sills	Tru-Jl	353.5°E, 29.5°N	261°E, 72°N	[5]
Draa Valley sills	Tru-Jl	353.5°E, 29.5°N	230.5°E, 65.5°N	[3]
Foum-Zguid dyke	Tru-Jl	353.5°E, 30.5°N	259°E, 58°N	[3]
Freetown Complex	Tru-Jl	347°E, 8.5°N	224.5°E, 81°N	[4]
Mateke Hills Complex	Jl	31°E, 22°S	259.5°E, 58.5°N	[17]
Atlas intrusives	Ju-Kl	354°E, 30°N	261.5°E, 53°N	[3]
Atlas intrusives	Kl	354°E, 32°N	251°E, 45°N	[6]
Mlanje Massif	Kl	35.5°E, 16°S	262°E, 60°N	[19]
Lupata volcanics	Kl	34°E, 16.5°S	259.5°E, 62°N	[20]
Moroccan sediments	Kl	353°E, 32°N	223.5°E, 75.5°N	[7]
Nubian iron ores	K	33°E, 24°N	204°E, 75°N	[8]
Nubian sandstone	Ku	34°E, 24.5°N	218°E, 63.5°N	[8]
Volcanics, Sicily	Ku	15°E, 36.5°N	223.5°E, 62°N	[9]
Kimberlite pipes	Ku	25°E, 29°S	237°E, 58°N	**
Mean Trl-Jl	$N = 17$	($k = 51.9, A_{95} = 4.9°$)	254.6°E, 69.6°N	
Mean Ju-Ku	$N = 9$	($k = 36.5, A_{95} = 8.6°$)	242.1°E, 62.8°N	
Mean all mesozoic	$N = 26$	($k = 42.4, A_{95} = 4.4°$)	249.5°E, 67.4°N	

* k is the precision parameter [27] and A_{95} the radius of the circle of 95% confidence about the mean.
** P.L. McFadden and D.L. Jones (to be published).

age very close to the Permo-Triassic boundary, they are the oldest yielding a palaeomagnetic pole concordant with other African Mesozoic poles. It appears that the palaeomagnetic pole with respect to Africa remained in essentially the same position throughout the whole of the Mesozoic.

4. Palaeoradius during the Mesozoic

There are now 26 pole positions for the Mesozoic determined from widely separated parts of Africa (Table 2, Fig. 2). They subdivide conveniently into two age groupings, one essentially Triassic (Early Triassic to Early Jurassic) and the other Cretaceous (Late Jurassic to Late Cretaceous). The mean poles for these two groups are given in Table 2, and they differ by only 7.9° which is not significant. This confirms the constancy of the African pole throughout the Mesozoic and provides very strong support for the geocentric dipole hypothesis. Recently it has been proposed [23] that there are small differences between Late Triassic to Early Jurassic results from southern Africa and the Atlas region of northwest Africa. However, when all the data are considered together the poles from both regions together form a cluster whose distribution is typical of poles sampled from regions of continental extent [24]. There is no reason to invoke a small

tectonic displacement for a part of Africa than there is in any other continent. The statistical scatter of results represents a combination of experimental errors, differences in time-averages for the various poles and minor tectonic displacements at *all* localities.

The widespread distribution of sampling sites from at least 35°N to 30°S in present latitude provides an excellent opportunity for testing hypotheses of earth expansion such as proposed by Carey [25]. One criticism levelled at palaeomagnetic estimates of the palaeoradius is that comparisons of results from widely separated localities are not valid because it can never be certain that the rocks from which the results are being compared have precisely the same age. This problem does not arise, however, because all Mesozoic results from Africa give indistinguishable palaeomagnetic poles.

The method set out by Ward [26] has been applied to the Triassic and Cretaceous groups of data separately. The method involves first finding the Fisher [27] mean of the site localities and then rotating all the palaeomagnetic parameters (site longitude and latitude, palaeomagnetic declination and inclination) to a new frame with the mean site position as the pole in spherical co-ordinates. In this frame of reference the grouping of palaeomagnetic poles (as measured by the Fisher [27] precision parameter k) for any hypothetical Earth radius R_a may be calculated. The best estimate of R_a (where $R_a = 1$ for the present radius) is that for which k is a maximum. The method automatically weights the data according to the distance apart of the sampling localities. The results of repeat calculations for different values of R_a are illustrated in Fig. 3. The Triassic data give a best estimate of $R_a = 1.08$ and the Cretaceous $R_a = 1.03$. If all the data are combined an overall Mesozoic value of $R_a = 1.08$ is determined.

The problem associated with the application of Ward's method is one of deciding on the error limits to be placed upon the estimates of palaeoradius. In the present instance some idea of the maximum errors involved may be determined by noting that the sampling localities (Fig. 2) are concentrated in northern Africa and southern Africa. They may be subdivided conveniently into a northern group including all localities above the present equator and a southern group with localities below the present equator. For each group the mean site location can be determined as in Ward's method and a mean palaeomagnetic pole calculated. A corresponding mean declination (D_m) and inclination (I_m) can then be calculated for each mean site location as listed in Table 3. When the data are arranged in this manner the mean site localities of the northern and southern groups lie close to a palaeomeridian. A close approximation to the palaeoradius may then be determined using the palaeomeridian method of Egyed [28] where:

$$R_a = \frac{\text{present angular separation of the sites}}{\text{past angular separation of the sites}}$$

For two sites situated on a palaeomeridian the past angular separation is simply the palaeolatitude difference between the sites. The error in R_a may then be determined from the error in the palaeolatitude difference which is simply calculated from the 95% circles of confidence for the mean palaeomagnetic poles for each group. The results of this calculation are summarized in Table 3. The palaeoradius estimates from this calculation are slightly higher than those given by Ward's method, but this is in the correct sense because slight departures from the palaeomeridian mean that the palaeolatitude difference between the sites is a slight underestimate of their past angular separation. The values listed in Table 3 therefore provide a very useful confirmation of the validity of Ward's method.

The errors in the palaeoradius calculated from Egyed's method and given in Table 3 are 12, 14 and 19% when the number of data points are 25, 17 and 9 respectively. The calculation, however, involves subdivision of the data into two roughly equal subsets, and the calculated overall mean error arises largely from combining the errors in these subsets where the numbers in each case are only half the total. In Ward's method all the data points are used in the single calculation suggesting that the resulting errors must be less than those assigned in Table 3. In the present instance therefore we suggest that the errors in the palaeoradius determined by Ward's method are about 80% of those given in Table 3. The results of the analysis of the Mesozoic palaeomagnetic data for Africa using Ward's method may then be summarized as follows, where the errors quoted are the 95% confidence limits:

Triassic palaeoradius $R_a = 1.08 \pm 0.15$

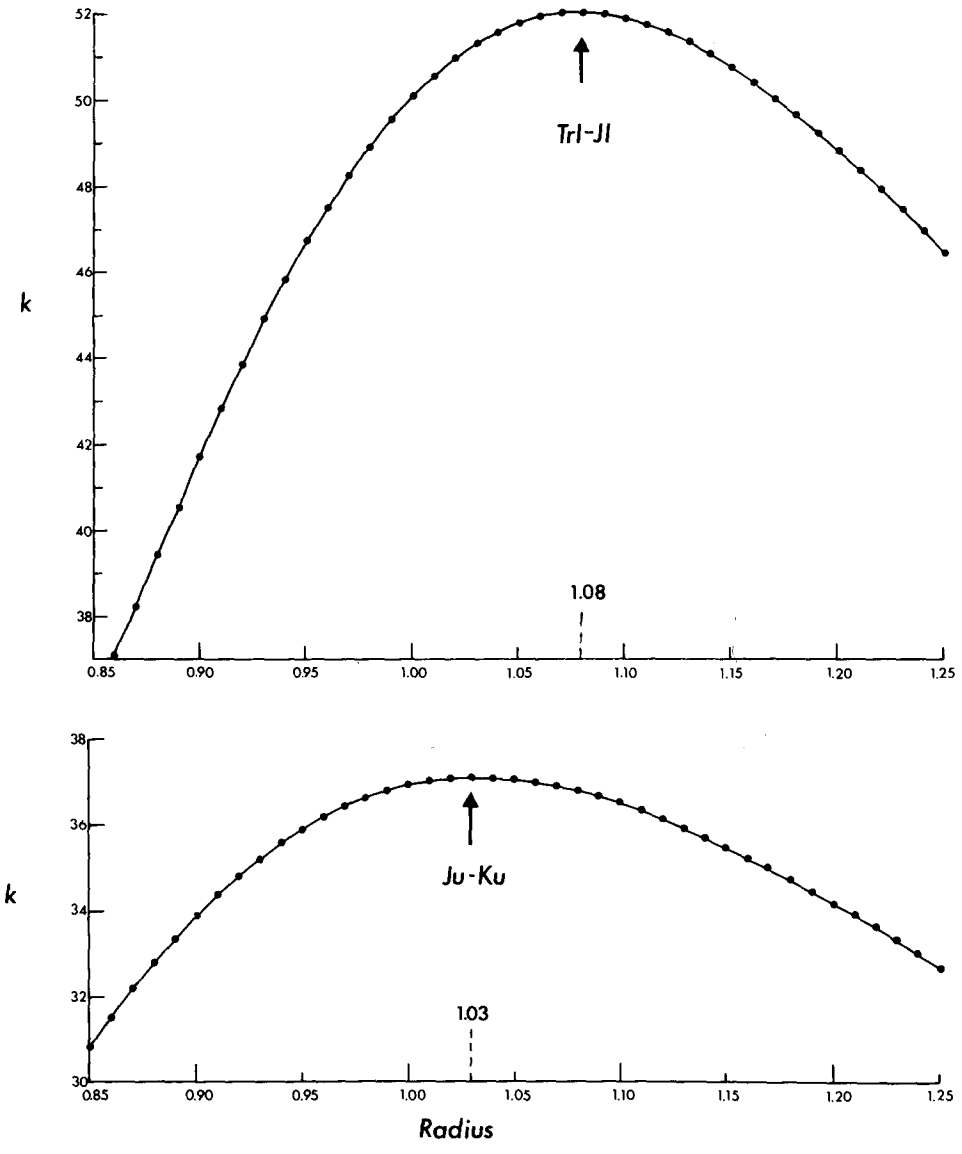


Fig. 3. Pole precision (k) as a function of palaeoradius (present radius = 1.0) for Triassic and Cretaceous data for Africa using Ward's [26] method.

Cretaceous palaeoradius $R_a = 1.03 \pm 0.19$

Overall Mesozoic palaeoradius $R_a = 1.08 \pm 0.13$

These values demonstrate that hypotheses of Earth expansion are very difficult to sustain.

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TABLE 3

Regional mean palaeomagnetic parameters for northern and southern Africa relating to estimates of the palaeoradius using Egyed's palaeomeridian method [28] *

Region	Mean site position	Angular difference (A)	D_m, I_m	Palaeolatitude difference (P)	Palaeoradius $R_a = A/P$
Triassic					
Northern Africa	28.0°N, 8.0°W	61.8°	343.5, +38.0	55.8° ± 10.0°	1.11 ± 0.20
Southern Africa	21.2°S, 30.7°E		337.0, −54.0		
Cretaceous					
Northern Africa	31.1°N, 10.8°E	55.4°	341.2, +22.0	52.3° ± 12.2°	1.06 ± 0.24
Southern Africa	20.5°S, 31.7°E		335.0, −60.0		
All Mesozoic					
Northern Africa	29.6°N, 0.2°W	60.1°	341.5, +32.5	53.7° ± 8.2°	1.12 ± 0.17
Southern Africa	21.0°S, 30.9°E		336.5, −55.5		

* Small departures of the line joining the mean site positions from the palaeomeridian give values slightly greater than their true value.

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